Package Surveillance System
User Manual

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1. SYSTEM OVERVIEW

The smart package surveillance device is an advanced security solution designed to protect parcels from theft. It is capable of detecting unauthorized movement or lifting of the package and immediately triggers a response. Once theft is detected, the system automatically captures images at regular intervals of 10 seconds to document the event. Simultaneously, it sends instant SMS alerts containing the GPS location of the package, ensuring real-time awareness. In addition, email notifications are sent with attached images and location details for further monitoring and evidence. The device also supports live tracking through a web interface, allowing users to continuously monitor the package's position. Powered by a battery, the system can operate efficiently for approximately 5 to 6 hours, making it suitable for short-term surveillance and transportation scenarios.

HOW IT WORKS

1. Place the device inside or under your package.
2. Send an SMS command to arm the system.
3. The device monitors movement using a load sensor.
4. When lifted it immediately:
    - Sends SMS alert with GPS location
    - Sends email with photo
    - Captures photos every 10 seconds
    - Updates GPS every 30 seconds
5. Track the package in real time using the web interface.

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2. WHAT YOU NEED

HARDWARE:

1. Raspberry Pi 4 with surveillance system installed
2. SIM7600X 4G LTE modem with SIM card
3. Load sensor
4. Camera module
5. GPS antenna
6. Battery pack (10,600 mAh)

SOFTWARE:

1. Tailscale mobile app
2. SMS capable phone
3. Gmail account

ACCOUNTS:

1. Tailscale account
2. Gmail account with App Password
3. Phone number for alerts
4. SIM card with SMS + 4G data

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3. INITIAL SETUP

STEP 1 – INSTALL TAILSCALE ON PHONE

Download from App Store / Play Store and sign in.

STEP 2 – INSTALL TAILSCALE ON RASPBERRY PI

- curl -fsSL https://tailscale.com/install.sh | sh

sudo tailscale up

- Authenticate and retrieve IP using:

tailscale ip -4

STEP 3 – CONFIGURE SYSTEM

Edit config.py
OWNER_PHONE = "+1234567890"
SENDER_EMAIL = "your.email@gmail.com"
SENDER_PASSWORD = "your_app_password"
RECIPIENT_EMAIL = "alert.email@gmail.com"
SIM_APN = "internet"

STEP 4 – START SYSTEM

cd /home/user1234/surveillance-system
python3 main.py

System will display:
PACKAGE SURVEILLANCE SYSTEM
Tracking URL
Camera test
Internet connection
Waiting for READY SMS command

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4. HOW TO USE THE SYSTEM

1. ARM SYSTEM

- Send SMS: READY
- Monitoring begins.

2. PLACE DEVICE
Place inside package ensuring load sensor detects lifting.

3. MONITOR PACKAGE
Options:
• Web interface
• SMS alerts
• Email alerts
Web interface:
http://100.116.112.37:5000
Shows GPS map and camera feed.

4. WHEN THEFT IS DETECTED
• SMS alert sent
• Email alert sent
• Photos captured every 10 seconds
• GPS updated every 30 seconds

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5. SYSTEM STATES

IDLE: System running but not monitoring.
ARMED: Monitoring package.
THEFT DETECTED: Photos and GPS tracking active.

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6. MONITORING YOUR PACKAGE

WEB INTERFACE
Open browser:
http://100.116.112.37:5000

Displays:
- System status
- GPS map
- Camera image
- Coordinates

SMS ALERT EXAMPLE
THEFT ALERT
Location link to Google Maps
Tracking link

EMAIL ALERT
Includes timestamp, location, and photo.

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7. SMS COMMANDS

READY – Arm system
DISARM – Disarm system
STATUS – Return system status

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8. TROUBLESHOOTING

CASE 1: SYSTEM DOES NOT RESPOND TO SMS
- Check SIM SMS enabled
- Verify phone number format
- Wait 10–15 seconds

CASE 2: WEB INTERFACE NOT ACCESSIBLE
- Ensure Tailscale connected
- Check IP address
- Ensure system running

CASE 3: GPS SHOWS 0.0
- Move device outdoors
- Wait 2–3 minutes

CASE 4: NO EMAIL ALERTS
• Check Gmail App Password
• Verify EMAIL_ENABLED = True

Case 5: CAMERA NOT WORKING
• Check cable
• Enable camera via raspi-config

Case 6: BATTERY DRAINS FAST
Increase intervals:
CAPTURE_INTERVAL = 20
GPS_UPDATE_INTERVAL = 60

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9. TECHNICAL SPECIFICATIONS

POWER

- Battery: Dual 21700 Li-ion cells (10,600 mAh)
- Voltage: 7.4V
- Runtime: 5–6 hours active

CAMERA

- Resolution: 1280×720
- Capture interval: 10 seconds

GPS

- Accuracy: 5–10 meters
- Update interval: 30 seconds

CONNECTIVITY

- 4G LTE modem
- SMS alerts
- Tailscale VPN

STORAGE

- 32GB SD card
- Auto-cleanup keeps last 100 images

SENSORS

- Load sensor with debounce protection

ADVANCED CONFIGURATION

1. **Change phone number:**

OWNER_PHONE = "+1234567890"

2. **Adjust capture intervals:**

CAPTURE_INTERVAL = 10
GPS_UPDATE_INTERVAL = 30

3. **Change image resolution:**

IMAGE_WIDTH = 1280
IMAGE_HEIGHT = 720

4. **Disable email alerts:**

EMAIL_ENABLED = False

SAFETY AND BEST PRACTICES

• Use provided USB-C charger
• Avoid extreme temperatures
• Charge battery fully

LEGAL
• Use only on your own packages
• Follow local privacy laws

PLACEMENT
• Device at bottom of package
• GPS antenna near top
• Camera unobstructed

SYSTEM MAINTENANCE

Regular checks:
• Clear old images
• Check battery health
• Test SMS commands
Restart system:
Ctrl + C
Then run
python3 main.py

SUPPORT AND CONTACT

Check manual first.
Common errors:

- SIM7600 not connected
- GPS fix not available
- Camera test failed
- Internet connection failed
- Email send error

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APPENDIX – FILE LOCATIONS

Main program:
/home/user1234/surveillance-system/main.py

Config:
/home/user1234/surveillance-system/config.py

Images:
/home/user1234/surveillance-system/images/

Latest image:
/home/user1234/surveillance-system/images/latest.jpg
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QUICK REFERENCE

SMS COMMANDS
READY → Arm
DISARM → Disarm
STATUS → Status

WEB INTERFACE
http://100.116.112.37:5000

BATTERY
Standby: 8–10 hours
Active: 5–6 hours
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